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Criteria development for sustainable construction manufacturing in Construction Industry 4.0

Theoretical and laboratory investigations

Sustainable practices

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Abstract

Purpose – This paper aims to present the sustainable performance criteria for 3D printing practices, while reporting the primarily computations and lab experimentations. The potential advantages for integrating three-dimensional (3D) printing into house construction are significant in Construction Industry 4.0; these include the capacity for mass customisation of designs and parameters for functional and aesthetic purposes, reduction in construction waste from highly precise material placement and the use of recycled waste products in layer deposition materials. With the ultimate goal of improving construction efficiency and decreasing building costs, applying Strand7 Finite Element Analysis software, a numerical model was designed specifically for 3D printing in a cement mix incorporated with recycled waste product high-density polyethylene (HDPE) and found that construction of an arched truss-like roof was structurally feasible without the need for steel reinforcements.

Design/methodology/approach – The research method consists of three key steps: design a prototype of possible structural layouts for the 3DSBP, create 24 laboratory samples using a brittle material to identify operation challenges and analyse the correlation between time and scale size and synthesising the numerical analysis and laboratory observations to develop the evaluation criteria for 3DSBP products. The selected house consists of layouts that resemble existing house such as living room, bed rooms and garages.

Findings – Some criteria for sustainable construction using 3DP were developed. The Strand7 model results suggested that under the different load combinations as stated in AS1700, the maximum tensile stress experienced is 1.70 MPa and maximum compressive stress experienced is 3.06 MPa. The cement mix of the house is incorporated with rHDPE, which result in a tensile strength of 3 MPa and compressive strength of 26 MPa. That means the house is structurally feasible without the help of any reinforcements. Investigations had also been performed on comparing a flat and arch and found the maximum tensile stress within a flat roof would cause the concrete to fail. Whereas an arch roof had reduced the maximum tensile stress to an acceptable range for concrete to withstand loadings. Currently, there are a few 3D printing techniques that can be adopted for this purpose, and more advanced technology in the future could eliminate the current limitation on 3D printing and bring forth this idea as a common practice in house construction.

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Originality/value – This study provides some novel criteria for evaluating a 3D printing performance and discusses challenges of 3D utilisation from design and managerial perspectives. The criteria are relied on maximum utility and minimum impact pillars which can be used by scholars and practitioners to measure their performance. The criteria and the results of the computation and experimentation can be considered as critical benchmarks for future practices.

Keywords 3D sustainable printing, 3DP, 3D printing, Structural design, Concrete, Arch roof, Sustainability, Additive manufacturing, Numerical modelling, High-density polyethylene, 3D sustainable building printing, Construction Industry 4.0, Construction processes

Paper type Research paper

1. Background review

The potential advantages for integrating three-dimensional (3D) printing (3DP) into house construction are significant in Construction Industry 4.0 (Ndlovu *et al.*, 2019; Baz *et al.*, 2020; Goel and Gupta, 2020); these include the capacity for mass customisation of designs and parameters for functional and aesthetic purposes, reduction in construction waste from highly precise material placement and the use of recycled waste products in layer deposition materials (Shakor *et al.*, 2020). Industry 4.0 refers to the Fourth Industrial Revolution, which is enabled by the use of the advanced technologies including 3DP into the production (Tjahjono *et al.*, 2017). While digital technology is advancing to facilitate the Construction Industry 4.0 implementation (Kontovourkis and Tryfonos, 2020; Shirowzhan *et al.*, 2020a, 2020b), there are limited attempts to develop criteria and investigate how to evaluate different aspects of the technology use, which hinders the adoption of technology (Kidwell, 2017; Sepasgozar *et al.*, 2018a, 2018b; Shirowzhan *et al.*, 2020a, 2020b).

The concept of sustainability in additive manufacturing (AM) has been investigated in different industries other than construction such as medical, aerospace and automotive products (Ryan *et al.*, 2017). However, the sustainable practice of digital AM in construction has not been fully and systematically investigated. This paper intends to investigate what factors of 3DP in building construction design associates with sustainability. The main focus of the paper is to highlight the sustainable aspect of this practice, so called 3D sustainable printing (3DSP). The paper uses the identified factors to develop the evaluation criteria for 3DSP. While previous studies presented different types of 3DP technologies and their advantages and limitations, there are no metrics or evaluation criteria to assess these practices from a sustainability perspective. The results of a series of experimentations, including theoretical analysis of a proposed building, and laboratory experimentations to identify challenges of production from managerial perspective. In both types of experimentations, we used brittle materials, including concrete or clay.

3DP technology has been identified as a trending technology, which is useful for developing the concept of construction automation (Tahmasebinia *et al.*, 2018; Zhang *et al.*, 2018; Mechtcherine *et al.*, 2019). A 3D printer was invented in the 1980s, but only in recent years, it has gained recognition as a useful technology for construction. Many recent studies have recommended this technology to be used for larger-scale production in construction (Tay *et al.*, 2017) and also suggested as one of required technologies of Construction Industry 4.0 (Dilberoglu *et al.*, 2017; Tjahjono *et al.*, 2017; Cakmakci, 2019; Schmidt *et al.*, 2019; Sheladiya *et al.*, 2019). Recent literature introduced several innovative concepts and 3DPs such as CONPrint3D (Mechtcherine *et al.*, 2019) and HINDCON (Hybrid INDustrial CONstruction) (Esnault *et al.*, 2019). At the same time, some others suggested this technology can significantly improve sustainability in construction by the use of recycled waste materials and increase net saving (Tay *et al.*, 2017). However, sustainable practices are not limited to using waste materials as resources, and other sustainability criteria have not been thoroughly explored. Ding (2008) discusses how sustainable practices can be evaluated by measuring four main criteria: maximising wealth, including profitability, measured as a benefit-cost ration, including ongoing maintenance and durability; maximising utility, being external benefits related to the wider community; whilst minimising resources and energy both embodied and operational over the full life cycle; and minimising impact to all environmental and heritage issues. While previous studies have principally focused on the value of efficiency and profitability, including input materials as resources, they have largely ignored maximising utility and minimising impact. This study intends to investigate how utility can be improved and how the impact in terms of producing waste can be minimised. Adopting Ding (2008) definition of sustainability, the term sustainable will be used in this paper to refer to the construction processes and the quality of the production, which affects the impact of the practice on environment. The process of 3D implementation in different scales needs to be analysed from the utility perspective and environmental impact.

In the theoretical part of the paper, the feasibility of 3DSP is examined to create structures capable of sustaining temperature conditions suitable for humans. The paper presents the analysis of a simplistic residential structure designed specifically for construction using a 3D printer. This research focuses on the process of prefabricating a fibre-reinforced concrete composite (FRC) in factory before transporting and installation to the allocated structure site. For pouring concrete on site, the curing of concrete under typical site conditions only able achieve around 65 per cent of its compressive strength in seven days and attain over 90 per cent of its compressive strength after 28 days (Kim *et al.*, 1998). Whereas precast concrete requires intensive transport and labour cost and limited design flexibility due to concrete being cast in factory. Regardless of which method is adopted, intensive labour work and time are needed. With the increasing cost of labour work and time required to finish construction work by hand, the conventional house construction method would only get more expensive over time. This paper intends to analyse the structural feasibility of designing a 3DSP to investigate the feasibility of building a 3DSP prototype in the future. The 3DSP is a useful technology for saving cost and time during the construction phase. Compared to traditional building construction, 3DSP offers significant reductions in labour costs and increased efficiency and speed in completing the construction, specifically due in this study to fast curing rates because of 3DSP material properties. When compared to traditional building construction, it saves the cost of labour and has a higher curing rate because of its material properties. At the time of writing this paper, no strict guidelines or related Australian standards exist to follow when designing for 3DSP. Currently the practitioners' knowledge must be relied upon to understand the 3D printer function and limitations. Nevertheless, current existing Australian standards for structural strength provide reliable guidance on designing for the structural integrity of the modelled structure. In particular, AS1170 (Standards, 2002) and AS3600 (Standards, 2009) are used for the design guidelines. This paper consists of a design prototype of possible structural layouts for the 3D-printed house.

The fibre reinforcement used in this project is recycled high-density polyethylene (rHDPE), which has received some interest in the literature (Christ *et al.*, 2015; Panda *et al.*, 2017; Shakor *et al.*, 2019a, 2019b). This fibre presents a viable alternative to conventional reinforcement for both stability and thermal expansion qualities, as outlined by various preceding studies (Pešić *et al.*, 2016). Moreover, rHDPE is relatively inexpensive compared to other alternatives while effectively achieving tensile reinforcement in 32 MPa concrete. In addition, thermal insulation identified as an increasing concern in the production of sections by this construction method. Gosselin *et al.* (2016) showed that thermal insulation efficiency of the 3D printed FRC section can be optimised by designing wall geometry to reduce thermal bridges existing between its inner and outer surface. The internal truss design present serves to optimise thermal insulation by minimising the contact between the inner and outer surface of the wall while maintaining structural performance. Moreover, overall internal thermal comfort is analysed in accordance with relevant government regulations. This paper will also assess the relevancy of economically viable thermal intervention systems that may be introduced to supplement natural insulation of the structure. This study presents a theoretical design that complies with AS1170 (Standards, 2002) for structural stability under a variety of load combinations. The model consists of 102 m^2 floor space dedicated to typical residential rooms, including living quarters and common spaces. The dimensions of the design maximise available floor space while also providing sufficient living space to accommodate for the general activities undertaken by its inhabitants.

Currently available 3DP technologies can be adopted for building construction, and this paper discusses the applications, advantages, limitations and future directions of 3DSP as an innovative and viable solution for affordable house construction. As the construction cost of a typical house keeps on increasing, it is crucial to find an innovative way to build house efficiently and cost effectively. 3D printer was invented back in the past century, but only in recent years (Colla *et al.*, 2013; Shakor *et al.*, 2017a, 2017b; Tay *et al.*, 2017; Shakor *et al.*, 2019a, 2019b), it gained its recognition in the market worldwide. The 3DP process contains a number of requirements associated with constructing concrete-based building designs. Firstly, this construction process is limited by the difficulty of incorporating steel reinforcements into the concrete matrix. This is because concrete is extruded through a nozzle via the 3D Printer and deposited layer by layer. The present investigation will integrate fibre reinforcement within the 3D-printed deposition material to improve the tensile strength of the concrete without disrupting the 3DSP process.

2. Research method

The research method consists of three key steps:

(1) design a prototype of possible structural layouts for the 3DSP;

(2) create 24 laboratory samples using a brittle material to identify operation challenges and analyse the correlation between time and scale size; and

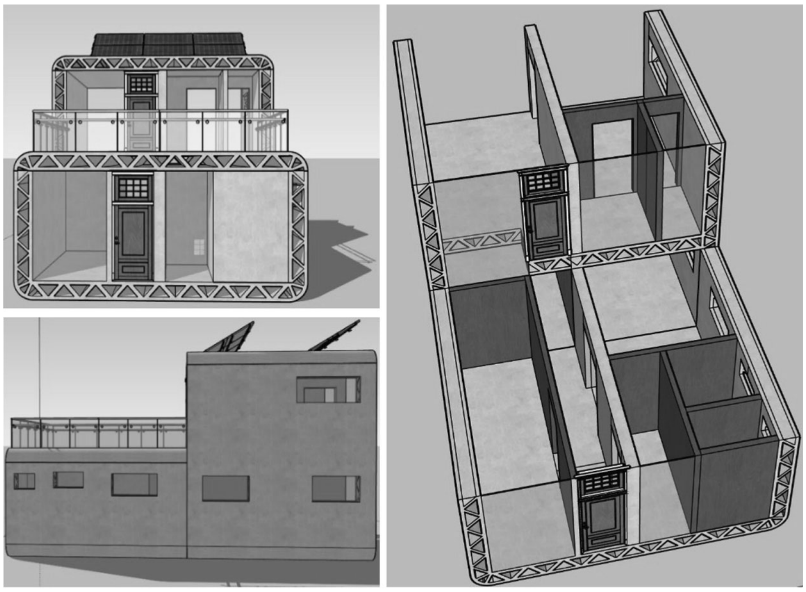
(3) synthesising the numerical analysis and laboratory observations to develop the evaluation criteria for 3DSP products.

The selected house consists of layouts that resemble existing house such as living room, bed rooms and garages. An analytical as well as a numerical model was built specifically designed for 3DSP, which consists of a truss-like roof spanning the entire structure. It is 22.3 m wide, 6 m high and 17 m deep structure; which would have glass panel covering the opening of the building for sun light to comes in.

The dimensions provide residents sufficient spatial area for daily activities. Numerical results from the finite element analysis software Strand7 are included to examine the structural strength of the design and in accordance with Australian standards for residential house construction. Structural analysis was also performed to investigate the advantage of arch roof against typical flat-slab roof for use in 3DP house. A novel analytical solution will be derived in line with the outcomes from the numerical analysis. The laboratory samples analysed over 24 experiments provided data for both future operational challenges that can be applied to on-site construction and any time increases and decreases at various scales. The correlation will be helpful in predicting the time per layer at larger scales.

Figure 1 shows the design of the proposed house measuring 12 m in length, 6 m in width and at 6.5 m in height. It includes truss-like elements such as walls, slab and roof. The proposed model is a structurally stable two-storey house with thermal insulation. The prefabricated construction method is adopted here to make the house more cost efficient and assure quality of work. The proposed material to be used is rHDPE, with compressive strength of 26 MPa and tensile strength of 3.06 MPa, which means the idea of steel replacement feasibility can be further investigated.

One limitation of 3DP concrete is the typical requirement of aggregate particles with a diameter of 20 mm or above, which cannot pass through the nozzle of a 3D printer with the consistency required. Another limitation of 3DP in construction is the need for using steel bar reinforcements. In this study, such steel bar reinforcements will be replaced by brittle materials, including clay or brittle plastic, rHDPE. According to Dym and Williams (2010), the addition of rHDPE into a concrete mix increases both the tensile strength and the serviceability of the concrete. Therefore, the inclusion of rHDPE in cement has a higher tensile strength than standard concrete and is beneficial as it eliminates the need for steel reinforcements. The differences in mixture ratio alters the compressive and tensile strength of concrete for different applications. The addition of rHDPE to the concrete mix can also control the durability of the concrete. The rHDPE microfibres, at 0.25 mm in diameter, are small enough for reliable and consistent material deposition by a 3D printer nozzle. With the addition of these fibres, the overall concrete elastic modulus becomes 25.2 GPa. The density, on the other hand, decreases to 2,380 kg/m3. The use of rHDPE has the added benefit of reducing environmental impact because it is commonly upcycled from other plastic waste products. The resulting cement mix also offers significant environmental benefits over steel



Note: The model consists of 102 m2 floor space including living quarters and common spaces

Figure 1.
Side and inside views of the proposed house conceptual design

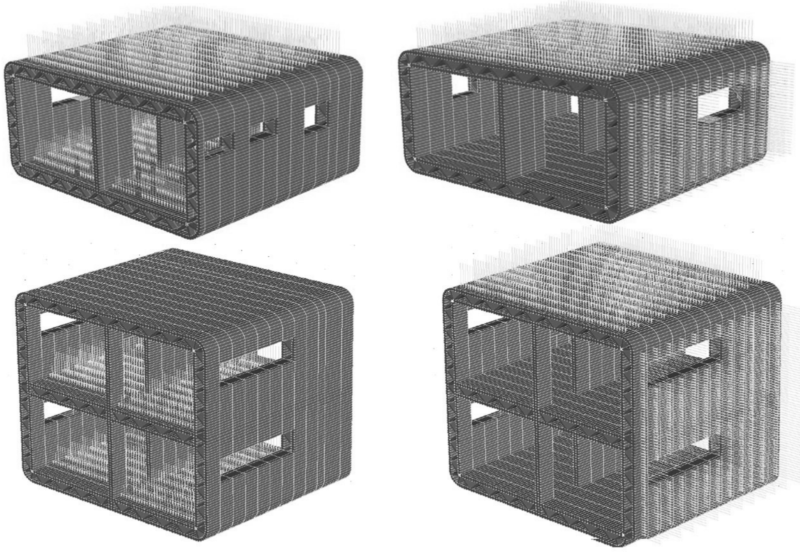
bar or mesh, as it is further a recyclable material when compared to reinforced concrete, which is generally difficult to recycle.

3. Structural composition analysis

Researchers such as Pešić *et al.* (2016) examined the idea of concrete polyethylene mix to use the use of 3DP techniques in concrete house constructions. Studies had shown the mixture with rHDPE can enhance the tensile strength of the concrete mix, which is viable to use within 3D-printed material. While being environmentally friendly and reducing carbon footprint for using recycled polyethylene, the increased overall tensile strength with mixture also compensate the lack of tensile reinforcement when using cement paste in 3DSP. Australian standards were used to check the structural response under load combination, to investigate the feasibility of using 3D house printing technology in house construction. As there is currently not a specific standard for 3DSP, this paper followed the AS1170 (Standards, 2002) series and AS3600 (Standards, 2009) as a guideline and design decision. One of the main challenges that the 3DP house needs to overcome is the lack of steel bar reinforcements within the structure. Reinforcements are common in modern structures; however, it is not feasible to use such reinforcement in a 3D printer. Therefore, tensile stress becomes the limiting factor. From the results if the finite element modelling on Strand7 for both structures, it can be concluded that this is a viable method of construction in terms of stability and thermal comfort.

The permanent load of the structure includes the slab and column self-weight, additional partitions, staircase weight and the implementation of a solar panel system on the non-habitable roof of the two-storey structure. These dead loads were calculated by accurately modelling the intended 3D-printed structure and applying a downward acting gravitational force. The density of the fibre reinforced concrete mix was taken as 2,380 kg/m^3. The live loads include imposed actions set as standard for various residential living spaces as outlined in AS1170. These loads were applied as normal pressure to the brick elements comprising the floor slab of the structure. The wind actions applied to the windward and leeward walls were calculated according to AS1170.2. The wind velocity for the region A2 is 45 m/s, and the design velocity for the location, cardinal direction and structural dimensions, is calculated at 35.483 m/s. Other parameters are calculated based on the AS1170.2 (Standards, 2002). Figures 2a and 2b illustrate the live and wind loads distribution on structures.

The mesh created below for this project consists of a series of Quad4 Plate elements designed to form the cross section of the structure. The cross section of the 3D-printed structure was created in Strand7 Finite Element Analysis Software without the aid of AutoCAD. The final resulting geometry of the cross section was dictated by testing parameters of this numerical analysis as well as constructability using the 3DP technique. The stability and thermal resistance of the structure determined the shape and composition of the mesh. This triangulated truss provided stability against lateral and horizontal loads by effectively distributing and transferring loads through the structure. Moreover, this helped to reduced self-weight of the slabs without sacrificing tensile/compressive capacities, which in turn allowed the implementation of a greater horizontal truss span. Another important benefit of the truss design is its efficiency in resisting thermal conductivity. This layout of internal and external walls separated by truss work has been shown to reduce the conductance of heat from outside sources by limiting thermal bridges transferring heat (Gosselin *et al.*, 2016). Moreover, this design increases the air pockets within the wall, which helps increase thermal resistance (Dos Santos and Mendes, 2009).



(a) (b)

Notes: (a) live loads; (b) wind loads varying from –0.831 (C_{fig} = –1.1) to 0.640 (C_{fig} = 0.84) KPa on windward and sidewalls respectively

Figure 2. Live and wind loads distribution on the structure

The stability of this structure was analysed using a 3D rendering of the concept design on Strand7. This model was subjected to permanent, imposed and wind actions in accordance with AS1170 (Standards, 2002) requirements to viably predict stability. A summary of the finite element model results is detailed below in Table I.

Fibre-reinforced concrete was identified early in the design stage of this project to be poor in tensile strength (relative to its compressive strength), which is characteristic of nominal concrete structures. Studies into rHDPE concrete show that the introduction of fibre reinforcement into the concrete mix increases the tensile strength from 1.6 to 3.08 MPa, while compressive strength of 32 MPa concrete increases to 34 MPa. Using

Model description	Permanent action (G)	Imposed action (Q)	Wind action (Wu)
Single-storey model			
Maximum DX displacement (m)	2.35×10^{-8}	5.2×10^{-9}	2.16×10^{-8}
Maximum DY displacement (m)	-6.22×10^{-8}	-1.71×10^{-8}	-4.67×10^{-8}
Maximum tensile stress (MPa)	0.380 (×)	0.082 (×)	0.056 (Y)
Maximum compressive stress (MPa)	0.652 (Y)	0.155 (Y)	0.088 (Y)
Double-storey model			
Maximum DX displacement (m)	4.45×10^{-8}	3.81×10^{-9}	1.29×10^{-7}
Maximum DY displacement (m)	-1.34×10^{-7}	-1.17×10^{-8}	-1.24×10^{-8}
Maximum tensile stress (MPa)	1.03 (Y)	0.066 (×)	0.191 (Y)
Maximum compressive stress (MPa)	2.32 (Y)	0.090 (Y)	0.204 (Y)

Table I. Finite element analysis of the model this as a guideline, it can be deduced from the table above that both structures can facilitate these load cases:

$$Maximum\ Tensile\ Stress\ induced\ (fti) = 1.03\,MPa < Tensile\ Strength\ (ftu) = 3.08\,MPa \quad (1)$$

The thermal analysis of this structure aims to predict the heat transfer generated by 3D-printed houses to ensure that it satisfies minimum thermal requirements. The thermal properties of rHDPE concrete were carried over from the reference project. This material has a specific heat of 880 J/kg/K and thermal conductivity of 1.37 W/m/K. As stated previously, studies have shown that this geometry has a multipurpose optimisation that not only contributes to structural stability but also thermal insulation (Gosselin *et al.*, 2016). This material alone is a poor insulator of heat based on its thermal resistivity (R factor), for reference, timber is:

$$R = \frac{t}{K} = \frac{0.61}{1.37} = 0.445 \quad (2)$$

It is necessary to understand that higher R factors imply a material is a good insulator. For comparison, other construction material R factors are:

$$220mm\ thick,\ Uninsulated\ Solid\ Brick:\ R = \frac{0.22}{1.2} = 0.18 \quad (3)$$

$$Insulated\ (with\ foil\ polyisocyanurate)\ Solid\ Brick:\ R = 0.18 + \frac{0.08}{0.022} = 3.82 \quad (4)$$

Given the method of calculating, the R factor (as seen above) only accounts for the conductivity of the material (not convection or radiation). Hence, it is better to evaluate the effectiveness of the walls in dissipating heat with thermal transmittance or U factor:

$$U = \frac{q}{\Delta T} \quad (5)$$

According to the Australian Bureau of Meteorology (BOM), the selected suburb of Hornsby experienced a maximum temperature of 42.7°C in the past two years of the experimentation time. This temperature (taken as a worst-case scenario) was applied as ambient convection to the outside face of the brick elements of the external walls and roof as well as the plate elements on both open ends of the structure. Conversely, the internal summer temperature of the house was assumed as 23°C (296.15 K), also applied as ambient convection. Additionally, solar radiation of 200 W/m3 was also applied as a heat source to the roof only of the structure. Lastly, a forced convection (due to wind gusts) of 170 J/sm^{2}K was applied to the external brick elements with natural convection of 12 J/sm^{2}K to the internal brick elements. The resulting model was generated using steady-state head analysis (Figure 3).

As it is illustrated by Figure 3, the temperature contour shows how maximum temperatures experienced outside the house can effectively dissipate between the external and internal walls. This geometrical design is able to maintain a relatively stable internal temperature, without much fluctuations within the structure.

Figure 4 demonstrates plotting the temperature along a single truss element between the walls shows how the truss maximises its length to dissipate heat. This temperature initially receives a large drop before gradually reducing as it transfers through the conducting material. This initially larger drop from 315.85 to 302.6 K may be the transfer of heat between the external wall and the truss where the thermal bridges have the largest area.

As it is illustrated by Figure 5, the temperature contour shows how maximum temperatures experienced outside the house can effectively dissipate between the external and internal walls. This geometrical design is able to maintain a relatively stable internal temperature, without much fluctuations within the structure. This temperature initially receives a large drop before gradually reducing as it transfers through the conducting material. This initially larger drop from 315.85 to 302.6 K may be the transfer of heat between the external wall and the truss where the thermal bridges have the largest area.

Increasing the thickness of a solid wall will improve the R factor and contribute to thermal insulation, however impractical. This truss geometry limits the thermal bridges available for heat transfer to the truss members and increases the required heat transfer distance of a 0.45 m thick wall to 0.61 m. This geometry maximises thermal insulation efficiency while minimising cost and material use. As a result, the internal face temperature of the brick elements only increased by approximately 0.29°C (Figure 6). Another possible concern surrounding thermal performance is the openings of this structure. Window openings on the side walls of the structure as well as the four large openings (two on each end) at the ends of the structure may be susceptible to higher heat transmittance than the FRC side walls (Figure 7). The heat flux through these regions may be significantly higher

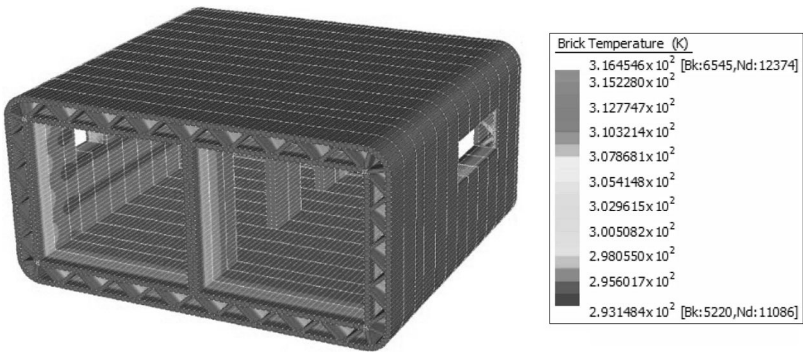


Figure 3.
Steady-state analysis of the model

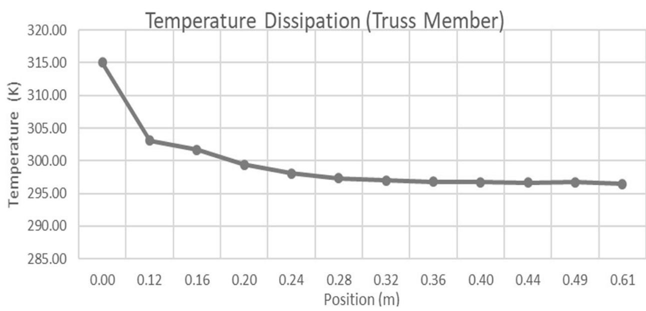


Figure 4.
Temperature versus length of the truss

due to higher U factors generally associated with single- or double-pane glass of approximately 1.6 W/m^2K (Aste *et al.*, 2009).

Due to the nature of the window opening, truss elements joining the external and internal members were different than the rest of the structure. Top and bottom flat truss members were implemented (Figure 7). The heat transfer distance is no longer angled and can only dissipate through 0.45 m of rHDPE concrete. However, steady-state heat analysis shows that this has minimal effect on the increase in wall temperature as only an additional temperature increase of 0.23°C and an overall temperature increase of 0.52°C resulted. Despite this, thermal resistance in these areas of increased heat flux is expected to lessen in magnitude once appropriate window panelling is installed (post-printing). Moreover, regions of particularly high heat transfer can also be subject to additional insulation by applying an insulative material such as foil polyisocyanurate or wool sheathing. To further support the results above, the U factor was calculated with equations (3) and (4). The outdoor temperature for a section of wall was averaged as 314.95 K, and the average indoor temperature for adjacent section was 296.85 K. The heat flux was calculated by inspecting the truss brick elements and averaging it to 14.1 W/m^2. Using steady-state heat numerical results as seen below:

$$U = \frac{1}{R} = \frac{1}{0.445} = 2.247 \tag{6}$$

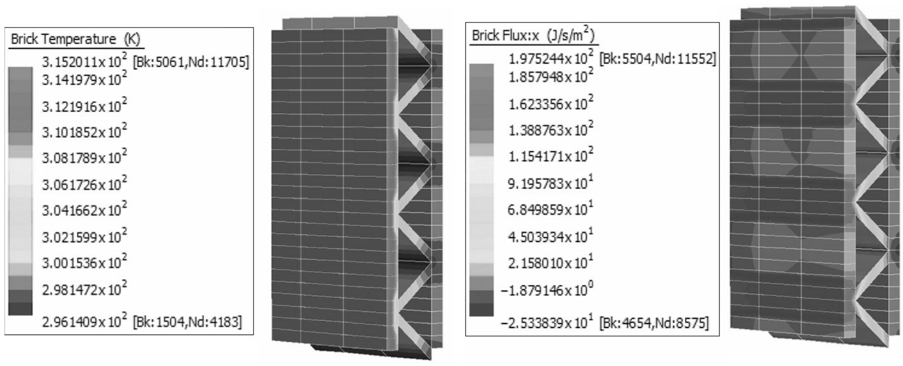


Figure 5.
Temperature variation in the truss

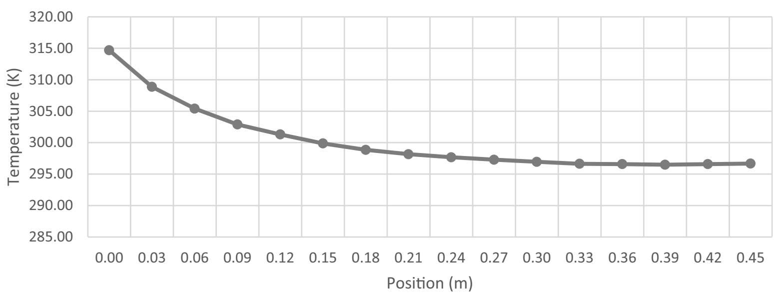


Figure 6.
Temperature dissipation through openings

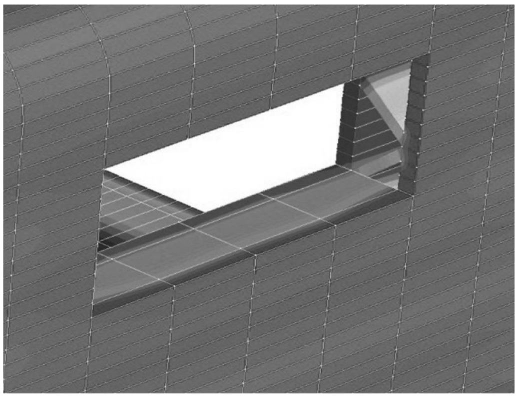


Figure 7. Heat dissipation through openings

$$U = \frac{q}{\Delta T} = \frac{14,1}{314.95 - 296.85} = \; 0.737 \tag{7}$$

For reference, Italian standards set U factor as 0.34 being the average thermal transmittance when built using conventional methods and implementing thermal insulation (Aste *et al.*, 2009). As the U factor is the inverse of the R factor, a lower U factor implies a better dissipation/resistance of heat transferring through the walls. It can be inferred from these results that the geometrical composition alone is effective in providing passive insulation to the internal structure. Hence, it can be deduced that this geometrical composition is definitively an improvement over a solid rHDPE concrete wall of the same thickness. Despite this, the insulation efficiency is limited by its thermal properties, which limits its range of improvement. Based on the calculated U factor, in comparison with typical residential wall transmittance, additional thermal insulation materials may need to be installed post-printing of the structure. Alternatively, thermal interventions such as cooling/heating systems may be implemented according to BASIX Thermal Protocols to provide assistance in stabilising internal ambient temperatures.

4. Analytical solution

An analytical method is developed to evaluate shear stress and strain distributions between the engaged surfaces throughout different joint layers by considering the beam theory method in different directions with respect to the different planes, where it can independently calculate shear forces between the different layers and shear strain as well as the curvature distribution along the different layers that have been extracted. The main concept to derive the following equations was extracted from Gilbert *et al.* (2016). According to Gilbert *et al.* (2016), the cross-sectional analysis is based on the assumption of the Euler–Bernoulli beam model. The strain distribution across the section can be calculated by $\varepsilon = \varepsilon_r - y \times \kappa$ where ε_r is the strain at the reference point (which can be determined at any point), y is the distance between the selected point and location of the neutral axis of the cross-section and κ is the curvature across the section in different strata layers. A vector can be introduced by $K(D)$, which will be included in the internal action N (axial forces) and M (internal moment). External loads, which might be due to the effect of the self-weight of the strata layers as well as the possible applied forces due to the vertical or horizontal displacement in the different layers, can induce the external axial force N_e and external moment M_e. The relationship between the internal and external actions can be presented by:

$$\varepsilon = \begin{bmatrix} \varepsilon_r \\ \boldsymbol{\kappa} \end{bmatrix} \tag{8}$$

$$r(\varepsilon) = \begin{bmatrix} N \\ M \end{bmatrix} \tag{9}$$

$$r_e = \begin{bmatrix} N_e \\ M_e \end{bmatrix} \tag{10}$$

$$r(\varepsilon) = r_e \quad (\text{This is the vector for strain}) \tag{11}$$

By considering the nonlinear interactions, the presented equations can be re-written by:

$$r(\boldsymbol{\varepsilon}^{(i+1)}) = r\left(\boldsymbol{\varepsilon}^{(i)}\right) + r_t\left(\boldsymbol{\varepsilon}^{(i)}\right) \times \Delta\boldsymbol{\varepsilon}^{(i)} = r_e \tag{12}$$

$$r_t\left(\boldsymbol{\varepsilon}^{(i)}\right) \times \Delta\boldsymbol{\varepsilon}^{(i)} = r_R^{(i)} \tag{13}$$

$$r_R^{(i)} = r_e - r\left(\boldsymbol{\varepsilon}^{(i)}\right) \tag{14}$$

$$\frac{\partial N\left(\boldsymbol{\varepsilon}_r^{(i)}, \boldsymbol{\kappa}^{(i)}\right)}{\partial \boldsymbol{\varepsilon}_r} \times \Delta\boldsymbol{\varepsilon}_r^{(i)} + \frac{\partial N\left(\boldsymbol{\varepsilon}_r^{(i)}, \boldsymbol{\kappa}^{(i)}\right)}{\partial \boldsymbol{\kappa}} \times \Delta\boldsymbol{\kappa} = N_R^{(i)} \tag{15}$$

$$N_R^{(i)} = N_e - N\left(\boldsymbol{\varepsilon}_r^{(i)}, \boldsymbol{\kappa}^{(i)}\right) \tag{16}$$

$$M_R^{(i)} = M_e - M\left(\boldsymbol{\varepsilon}_r^{(i)}, \boldsymbol{\kappa}^{(i)}\right) \tag{17}$$

All the equations can be re-presented in a matrix format:

$$r_t\left(\boldsymbol{\varepsilon}^{(i)}\right) = \begin{bmatrix} \dfrac{\partial N\left(\boldsymbol{\varepsilon}_r^{(i)}, \boldsymbol{\kappa}^{(i)}\right)}{\partial \boldsymbol{\varepsilon}_r} & \dfrac{\partial N\left(\boldsymbol{\varepsilon}_r^{(i)}, \boldsymbol{\kappa}^{(i)}\right)}{\partial \boldsymbol{\kappa}} \\ \dfrac{\partial M\left(\boldsymbol{\varepsilon}_r^{(i)}, \boldsymbol{\kappa}^{(i)}\right)}{\partial \boldsymbol{\varepsilon}_r} & \dfrac{\partial M\left(\boldsymbol{\varepsilon}_r^{(i)}, \boldsymbol{\kappa}^{(i)}\right)}{\partial \boldsymbol{\kappa}} \end{bmatrix} \tag{18}$$

$$\Delta\boldsymbol{\varepsilon}^{(i)} = \begin{bmatrix} \Delta\boldsymbol{\varepsilon}_r^{(i)} \\ \Delta\boldsymbol{\kappa}^{(i)} \end{bmatrix} \tag{19}$$

$$\text{(Changing strain and curvature)} \quad r_R^{(i)} = \begin{bmatrix} N_R^{(i)} \\ M_R^{(i)} \end{bmatrix} \tag{20}$$

The partial derivatives of N and M with respect to $\boldsymbol{\varepsilon}_r$ and $\boldsymbol{\kappa}$ can be rearranged in a more practical form, recalling the definitions of internal actions as:

$$\frac{\partial N\left(\boldsymbol{\varepsilon}_r^{(i)}, \boldsymbol{\kappa}^{(i)}\right)}{\partial \boldsymbol{\varepsilon}_r} = \int \frac{\partial \boldsymbol{\sigma}}{\partial \boldsymbol{\varepsilon}_r} dA \tag{21}$$

$$\frac{\partial N\left(\boldsymbol{\varepsilon}_r^{(i)}, \boldsymbol{\kappa}^{(i)}\right)}{\partial \boldsymbol{\kappa}} = \int \frac{\partial \boldsymbol{\sigma}}{\partial \boldsymbol{\kappa}} dA \tag{22}$$

$$\frac{\partial M\left(\boldsymbol{\varepsilon}_r^{(i)}, \boldsymbol{\kappa}^{(i)}\right)}{\partial \boldsymbol{\varepsilon}_r} = -\int y \frac{\partial \boldsymbol{\sigma}}{\partial \boldsymbol{\varepsilon}_r} dA \tag{23}$$

$$\frac{\partial M\left(\boldsymbol{\varepsilon}_r^{(i)}, \boldsymbol{\kappa}^{(i)}\right)}{\partial \boldsymbol{\kappa}} = -\int y \frac{\partial \boldsymbol{\sigma}}{\partial \boldsymbol{\kappa}} dA \tag{24}$$

where the values of the stress depend on the constitutive models adopted for the materials and on the magnitude of the strain:

$$\frac{\partial N\left(\boldsymbol{\varepsilon}_r^{(i)}, \boldsymbol{\kappa}^{(i)}\right)}{\partial \boldsymbol{\varepsilon}_r} = \int \frac{\partial \boldsymbol{\sigma}}{\partial \boldsymbol{\varepsilon}_r} 2\mathrm{d}A = \int \frac{\partial \boldsymbol{\sigma}}{\partial \boldsymbol{\varepsilon}} \times \frac{\partial \boldsymbol{\varepsilon}}{\partial \boldsymbol{\varepsilon}_r} dA = \int \frac{\partial \boldsymbol{\sigma}}{\partial \boldsymbol{\varepsilon}} \times \frac{\partial(\boldsymbol{\varepsilon}_r - y \times \boldsymbol{\kappa})}{\partial \boldsymbol{\varepsilon}_r} dA = \int \frac{\partial \boldsymbol{\sigma}}{\partial \boldsymbol{\varepsilon}} dA \tag{26}$$

$$\frac{\partial N\left(\boldsymbol{\varepsilon}_r^{(i)}, \boldsymbol{\kappa}^{(i)}\right)}{\partial \boldsymbol{\kappa}} = \int \frac{\partial \boldsymbol{\sigma}}{\partial \boldsymbol{\kappa}} dA = \int \frac{\partial \boldsymbol{\sigma}}{\partial \boldsymbol{\varepsilon}} \times \frac{\partial(\boldsymbol{\varepsilon}_r - y \times \boldsymbol{\kappa})}{\partial \boldsymbol{\kappa}} dA = -\int y \times \frac{\partial \boldsymbol{\sigma}}{\partial \boldsymbol{\varepsilon}} dA \tag{25}$$

$$\frac{\partial M\left(\boldsymbol{\varepsilon}_r^{(i)}, \boldsymbol{\kappa}^{(i)}\right)}{\partial \boldsymbol{\varepsilon}_r} = -\int y \times \frac{\partial \boldsymbol{\sigma}}{\partial \boldsymbol{\varepsilon}_r} dA = -\int y \times \frac{\partial \boldsymbol{\sigma}}{\partial \boldsymbol{\varepsilon}} \times \frac{\partial(\boldsymbol{\varepsilon}_r - y \times \boldsymbol{\kappa})}{\partial \boldsymbol{\varepsilon}_r} dA = -\int y \times \frac{\partial \boldsymbol{\sigma}}{\partial \boldsymbol{\varepsilon}} dA \tag{27}$$

$$\frac{\partial M\left(\boldsymbol{\varepsilon}_r^{(i)}, \boldsymbol{\kappa}^{(i)}\right)}{\partial \boldsymbol{\kappa}} = -\int y \times \frac{\partial \boldsymbol{\sigma}}{\partial \boldsymbol{\varepsilon}_r} dA = -\int y \times \frac{\partial \boldsymbol{\sigma}}{\partial \boldsymbol{\varepsilon}} \times \frac{\partial(\boldsymbol{\varepsilon}_r - y \times \boldsymbol{\kappa})}{\partial \boldsymbol{\kappa}} dA = \int y^2 \times \frac{\partial \boldsymbol{\sigma}}{\partial \boldsymbol{\varepsilon}} dA \tag{28}$$

$$\sigma = E \times \varepsilon \quad \text{for} \quad |\varepsilon| \leq \varepsilon_p \text{ (elastic strain)} \tag{29}$$

$$\sigma = f_p \quad \text{for} \quad |\varepsilon| > \varepsilon_p \text{ (plastic strain)} \tag{30}$$

$$\frac{\partial \sigma}{\partial \varepsilon} = \frac{\partial (E \times \varepsilon)}{\partial \varepsilon} = E \quad \text{for} \quad |\varepsilon| \leq \varepsilon_p \text{ (elastic strain)} \tag{31}$$

$$\frac{\partial \sigma}{\partial \varepsilon} = \frac{\partial (f_p)}{\partial \varepsilon} = 0 \quad \text{for} \quad |\varepsilon| > \varepsilon_p \text{ (plastic strain)} \tag{32}$$

$$N\left(\varepsilon_r^{(i)}, \boldsymbol{\kappa}^{(i)}\right) = \int \sigma dA = \sum_{j=1}^{n_j} \sigma\left(y_j, \varepsilon_r^{(i)}, \boldsymbol{\kappa}^{(i)}\right) \times A_j \tag{33}$$

$$M\left(\varepsilon_r^{(i)}, \boldsymbol{\kappa}^{(i)}\right) = -\int y \sigma dA = -\sum_{j=1}^{n_j} y_j \times \sigma\left(y_j, \varepsilon_r^{(i)}, \boldsymbol{\kappa}^{(i)}\right) \times A_j \tag{34}$$

$$\frac{\partial N\left(\varepsilon_r^{(i)}, \boldsymbol{\kappa}^{(i)}\right)}{\partial \varepsilon_r} = \int \frac{\partial \sigma}{\partial \varepsilon} dA = \sum_{j=1}^{n_j} \frac{\partial \sigma\left(y_j, \varepsilon_r^{(i)}, \boldsymbol{\kappa}^{(i)}\right)}{\partial \varepsilon} \times A_j \tag{35}$$

$$\frac{\partial N\left(\varepsilon_r^{(i)}, \boldsymbol{\kappa}^{(i)}\right)}{\partial \boldsymbol{\kappa}} = -\int y \times \frac{\partial \sigma}{\partial \varepsilon_r} dA = -\sum_{j=1}^{n_j} y_j \times \frac{\partial \sigma\left(y_j, \varepsilon_r^{(i)}, \boldsymbol{\kappa}^{(i)}\right)}{\partial \varepsilon} \times A_j \tag{36}$$

$$\frac{\partial M\left(\varepsilon_r^{(i)}, \boldsymbol{\kappa}^{(i)}\right)}{\partial \varepsilon_r} = -\int y \times \frac{\partial \sigma}{\partial \varepsilon_r} dA = -\sum_{j=1}^{n_j} y_j \times \frac{\partial \sigma\left(y_j, \varepsilon_r^{(i)}, \boldsymbol{\kappa}^{(i)}\right)}{\partial \varepsilon} \times A_j \tag{37}$$

$$\frac{\partial M\left(\varepsilon_r^{(i)}, \boldsymbol{\kappa}^{(i)}\right)}{\partial \boldsymbol{\kappa}} = \int y^2 \times \frac{\partial \sigma}{\partial \varepsilon} dA = \sum_{j=1}^{n_j} y_j^2 \times \frac{\partial \sigma\left(y_j, \varepsilon_r^{(i)}, \boldsymbol{\kappa}^{(i)}\right)}{\partial \varepsilon} \times A_j \tag{38}$$

The discussed practical equations are the equations that were comprehensively used by Strand7 to evaluate the stability analysis based on the different load combinations, including dead load, live load as well as the wind loads in different directions. By presenting and evaluating equations (8)-(38), a sound concept of the computational procedure was adequately indicated.

5. Laboratory experimentations and challenges

To find out the practical challenges of 3DSP, two models were created. Due to limitations of the used 3D printing, clay was selected as another type of brittle material to investigate practical deficiencies and challenges of the 3DSP procedure. Figure 8 shows two different forms created by a 3DSP at different small scales. The laboratory-based practices were conducted to identify the practical possible challenges of 3DSP, which might be existing on the main construction site.

Previous studies have mainly discussed the details of the equipment itself and possibility of producing larger scales of buildings (Tay *et al.*, 2017). Dissimilar to previous practices, this experimentation was carried out to identify key operational challenges of producing building elements aiming to be applied to large scale production. Using a PotterBot Scara XLS-2 clay printer, a total of 24 samples in different scales varying from 100 to 210 per cent of the designed model were produced in the laboratory. The samples of both square and rectangle shapes were created in the laboratory for careful observation of the production process from operation management perspective. The original size of the square and rectangle models are $13 \times 13 \times 0.6\,cm^3$ and $13.2 \times 6.8 \times 2.1\,cm^3$, respectively. Each model was taken 8.23 and 5.29 min, respectively, to be completed at the original 100 per cent scale size. Figure 9 shows the relationship between the scale in percentage and the time per layer. It shows that the time per layer is linear; however, the time per layer sharply increases in the scale changes of 120 and 140 per cent. While the change pattern of both lines in the figure is still linear, the line slope increases differently. Figure 5 also shows that the total time is affected by the scale. While the total time of producing samples increases, the slope of the changes suddenly changed between 120 and 160 per cent. These scatters provide useful insight into the time and production process. Only two components of the proposed design were used in the laboratory. Future studies can continue this experimentation by considering more samples and different components to show whether the relationship between size and time presents any specific pattern. This can be used as a prediction model for estimating the time of the building production.

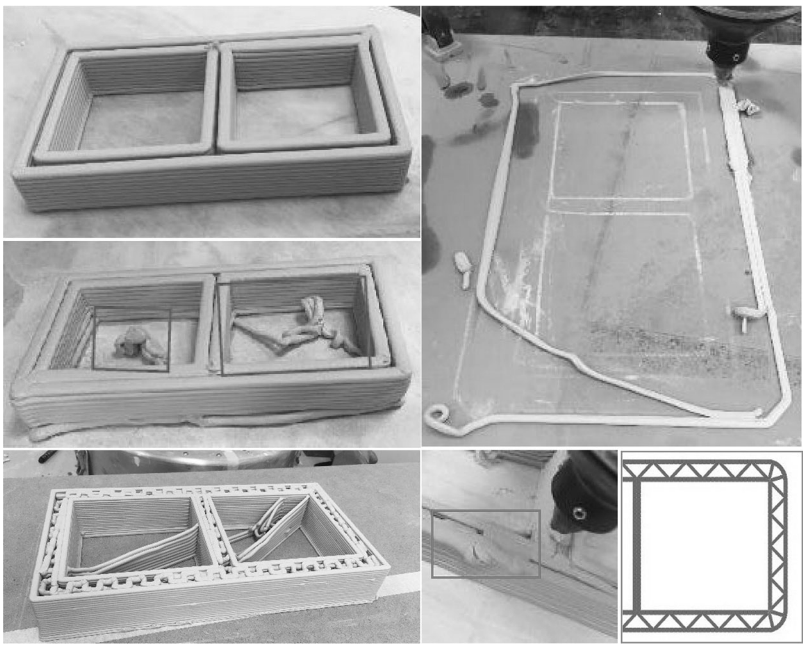


Figure 8.
Samples of produced models and defects including the structural element model

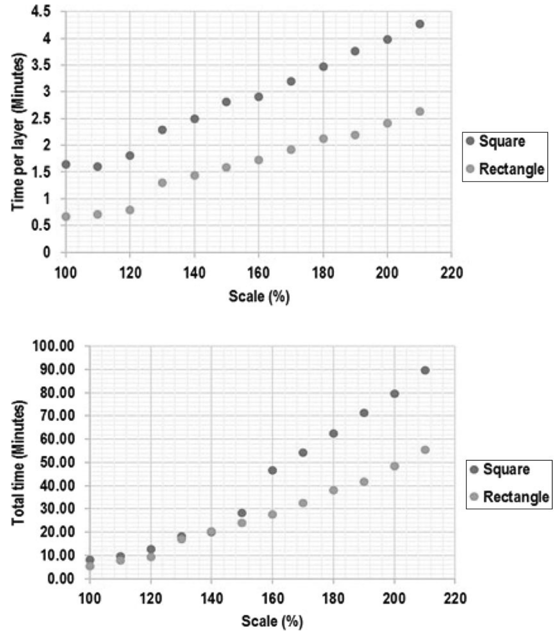


Figure 9.
The scatter showing the relationship between the scales of the samples ranging from 100 to 210 per cent

6. The 3D sustainable printing criteria development and discussion

This section synthesises the findings and observations discussed in the previous sections and presents the 3DSP criteria discussing the lessons learnt from both practical analysis of producing samples and the theoretical analysis of the concrete sample. The potential advantages of the 3DP concept is its capacity for mass customisation of designs and parameters for functional and aesthetic purposes, reduction in construction waste from highly precise material placement and the use of recycled waste products in layer deposition materials. The contribution of this paper is two fold:

(1) theoretical analysis of the proposed model for using cement materials; and
(2) developing criteria for evaluating the performance of the volumetric modules produced by 3DSP.

From a managerial perspective, the experimentations show that there are three main challenges for using the current technology for large samples.

The material change is a challenge, and the system works well when it is a single material (Duballet *et al.*, 2017; Kazemian *et al.*, 2017). However, construction managers need to use two or more materials for some parts of buildings. In these cases, hybrid systems are required to be able to produce the target design. The experimentation shows that materials should be appropriately chosen to maintain the design shape.

From design perspective, the experimentation shows that the normal drawings for traditional building construction may not be efficient. Different types of drawings, information modelling and coding strategies were tried in terms of optimisation of paths and connection of lines, which leads the machine arm to move from different angles. The machine movements and the arm paths affect the speed of producing the object. In addition, it may produce unwanted lines or materials as shown previously.

6.1.3D sustainable printing evaluation criteria

One of the major barriers of 3DSP is programming the machine in an efficient way. It is not enough to draw the building plan in a convenient way. The drawing should be revised in a way the 3DSP robot moves efficiently. In addition, extra motions cause the delivery of unwanted materials by the robot. As a future work, the motion programming should be investigated in different building cases. In addition, it should be clarified whether it is possible to add a desktop to the 3DSP machine and command something to the machine when it is working in the construction site. In this practice, we used G-code for the 3DSP robot interpreter. This is the common control language to move nozzle and control its travelling speed and position.

Using several 3DSP robots concurrently is a complicated practice because the motions should be programmed in a way to avoid collisions (Zhang *et al.*, 2018) and oscillation (Shakor *et al.*, 2017a, 2017b). In addition, the localisation should be highly precise to ensure all printed objects are aligned. Table II illustrates the relationship between the possible maximum utility as well as minimising the induced impact criterions. In the current study, some of the factors, including standards, drawing and coding, shape, stability, structural integrity and strength, as well as deformability are investigated.

The criteria development was designed because it is applicable to different 3DSP practices, as it offers a way to measure its performance in terms of utility and impact. Tay *et al.* (2017) discuss that there is a need to improve the understanding of the economics and environmental aspects of 3DSP practices to achieve maximum net savings. The criteria were developed based on Ding's (2008) concept of "maximising utility" and "minimising impact". However, the criteria can be extended further to cover economic viability, environmental

Criteria	Maximise utility	Minimise impact
Standards in Construction Industry 4.0	Develop local standards for design and construction	Implement local standards with less tolerance
Drawing and coding	Optimise the arm path; changeability of the extrusion nozzle diameter for different materials (Bodaghi *et al.*, 2017)	Use multiple and local materials
Shape stability (Zhou and Sheiko, 2016)	Finding the self-stability condition	Reducing the number of non-structural members to induce the overall self-weight of the structure
Material (Hiller and Lipson, 2010)	Percentage of fibres, local material or multiple material optimise the efficiency of the work, e.g. concrete polyethylene mix	The fabrication process should be compatible with several materials in a single build process
Structural integrity and strength	Fibres increases the stability	$\frac{\partial M\left(\boldsymbol{\varepsilon}_r^{(i)}, \boldsymbol{\kappa}^{(i)}\right)}{\partial \boldsymbol{\kappa}} = \int y^2 \times \frac{\partial \boldsymbol{\sigma}}{\partial \boldsymbol{\varepsilon}} dA = \sum_{j=1}^{n_j} y_j^2 \times \frac{\partial \sigma\left(y_j, \boldsymbol{\varepsilon}_r^{(i)}, \boldsymbol{\kappa}^{(i)}\right)}{\partial \varepsilon} \times A_j$
Deformation (Tay *et al.*, 2019)	Maximises the overall ductility of the structures	Mixture of the input materials

Table II. The evaluation criteria for 3DSP in larger-scale practices, including factors affecting utility and impact

impact and social effects. Sala *et al.* (2015) discussed more factors to be investigated such as stakeholders' involvement, scalability, "strategicness" and "integratedness". Other limitations of this research are related to 3DSP techniques, technological challenges for developing nations, large-capital cost preventing easy adoption. These factors are critical to small and medium enterprises (SMEs), as they follow different roles than pioneers in the industry (Hong *et al.*, 2016; Sepasgozar and Loosemore, 2017).

6.2 Theoretical implications

3D printed material deposition as a construction technique has only recently begun to be tested at build scale and minimal industrial experience or practical implementation exists. Researchers such as Pešić *et al.* (2016) examined the idea of concrete polyethylene mix to utilise the use of 3DSP techniques in concrete house constructions. Studies have shown the mixture with rHDPE can enhance the tensile strength of a concrete mix, which is viable to use within 3D-printed material. Furthermore, 3DSP with rHDPE can be more environmentally sustainable in reduction of the carbon footprint by using recycled waste material in place of steel reinforcements whilst maintaining sufficient tensile strength. Australian standards were applied to evaluate the structural response under load combination, to investigate the feasibility of using 3D house printing technology in house construction. As there is currently not a specific standard for 3DSP, this paper followed the AS1170 (Standards, 2002) series and AS3600 (Standards, 2009) as a guideline and in making design decisions. This paper suggests some criteria for evaluating the performance of 3DSP technology from the sustainability perspective. This can be used by other scholars to evaluate the performance of their practice in terms of utility and impact.

6.3 Managerial challenges and implications

There are several barriers to use 3DSP in practice. The major obstacle is the limitation of the size of the object printed by the 3DP robots (Zhang *et al.*, 2018; Mechtcherine *et al.*, 2019) and also limitations in nozzle, materials and pumping (Tay *et al.*, 2017; Ngo *et al.*, 2018). The current papers show that the 3DSP robot should be taller than the building structure. For example, they use gantry-based systems, including an arm relying on a large framework to support a nozzle (Zhang *et al.*, 2018). The papers showed a successful practice; however, the barrier of size rate of the 3DP to the building object still remains as the main barrier of 3DSP adoption. While technology adoption is investigated as series of activities taken by an organisation to make a decision to use a new technology (Sepasgozar *et al.*, 2016; Sepasgozar and Davis, 2018; Sepasgozar *et al.*, 2018a, 2018b), specific barriers of 3DP adoption are also discussed in the literature (Buswell *et al.*, 2018; Esnault *et al.*, 2019).

Motion planning is another major barrier of 3DSP is programming the machine in an efficient way. It is not enough to draw the building plan in the traditional two dimensionally documented plans, elevations and sections. The drawing must be revised and a 3D digital environment so that the 3DSP robot motion can be programmed efficiently. Any unnecessary motion by the 3 D printer nozzle can lead to deposition of unwanted material. Research has also found that currently available material deposition technologies do not have effective pause and resume functions within nozzle deposition design, and this required further investigation and development for both the hardware programming and material properties that will allow for predictable pause-resume material deposition. As for future studies, motion programming should be investigated in different building cases. In addition, it should be clarified whether it is possible to integrate a user-friendly software platform to the 3DP machine to allow instant modifications and adjustments when working on the construction site.

In our experiments, we used G-code to numerically control the 3DSP robot. G-code is the common control language for three-axes motion control that directs the travel on the X, Y and Z axes as well as the speed. Using the A PotterBot Scara XLS-2 clay printer, the speed of material deposition could not be controlled by the G-code generated, and this aspect requires further research to be applied to full-scale buildings on construction sites. However, other subtractive computer numerically controlled (CNC) technologies such as CNC routers have incorporated spindle rotation speeds into G-code, and this should be equally applicable to AM. Subtractive technologies, which cut away material from stock, has a history of development from the 1950s and therefore has improved the precision and efficiency to a higher level than AM. 3DP, and more broadly, AM, requires further development to its hardware, software and material properties to reach a similar adoption by the construction industry as the manufacturing industry has embraced subtractive CNC technologies.

7. Conclusions

This study aimed to present two main criteria for evaluating a sustainable practice such as how utility and the advantages that relate to the wider community can be improved and how the impact in terms of producing waste can be minimised. This paper assesses the viability of the novel concept of using 3DSP technology for increasing sustainability practices. The potential advantages for integrating 3DSP into house construction are significant; these include the capacity for mass customisation of designs and parameters for functional and aesthetic purposes, reduction in construction waste from highly precise material placement and the use of recycled waste products in layer deposition materials. With the ultimate goal of achieving the two criteria, applying Strand7 Finite Element Analysis software, a numerical model was designed specifically for 3DSBP in a cement mix incorporated with recycled waste product rHDPE and found that construction of an arched truss-like roof was structurally feasible without the need for steel reinforcements. The lab sizes prototypes were manufactured based on the destined numerical model by using a 3DSP approach.

The Strand7 model results suggested that under the different load combinations as stated in AS1700, the maximum tensile stress experienced is 1.70 MPa and maximum compressive stress experienced is 3.06 MPa. The cement mix of the house is incorporated with rHDPE, which result in a tensile strength of 3 MPa and compressive strength of 26 MPa. One of the main challenges that 3DP of houses needs to overcome is the elimination of steel bar reinforcements within the structure.

The contribution of this paper is twofold:

(1) developing criteria for evaluating the performance of the volumetric modules produced by 3DSP technologies; and

(2) theoretical analysis of proposed model for using cement materials.

In addition, this paper discusses the lessons learnt from both practical analysis of producing samples and the theoretical analysis of the concrete specimen.

This paper assesses the viability of the novel concept of using 3DSP technology for increasing sustainability practices. This type of construction techniques had only begun recently, and there exists no extensive industrial experience and practice in the past. The results show that the maximum tensile stresses induced in the structure from principle loading cases do not exceed the maximum tensile strength of the rHDPE concrete composite. Moreover, the geometrical composition of the exterior of the structure (truss framework) assists in creating passive insulation against heat ingress. Despite this, more research is required into thermal properties, especially in transient heat analysis. From the Strand7 results, it can be seen clearly that the structure would be able to sustain different load combinations without failing. Further investigation and development on the 3SDP technology should be made based on the criteria developed in the future, to provide society a more effective and sustainable method of building houses. The 3DSP is still in early stages of development and requires more investigations and advance detailing to be used in Construction Industry 4.0. To implement it in bulk, the system needs to be upgraded and constant learning process is required at both laboratory and practical case study levels.

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